Bank Group Credit Dynamics:

Regional, Sectoral, and Temporal Analysis

1. Project Overview:

- Bank Group Credit Dynamics: Regional, Sectoral, and Temporal Analysis examines how credit extended by Scheduled Commercial Banks (SCBs) in India varies across bank groups, population segments (rural, urban, metropolitan), and credit sectors (agriculture, industry, housing, personal) over time.
- The project cleans and harmonizes RBI groupwise credit tables into a tidy fact table, builds an interactive Power BI dashboard to visualize trends and concentrations, and produces evidence-based recommendations for policymakers and bank strategists.

1.1. Primary Objective

Assess temporal patterns and cross-sectional differences in SCB credit to identify where credit growth is concentrated, which bank groups drive sectoral shifts, and where regional imbalances or concentration risks exist.

Analytical objectives

- To compute total credit and growth rates by **bank group**, **region (rural/urban/metropolitan)**, **sector**, and **year**.
- To quantify how much of overall credit growth is driven by changes in volume versus changes in composition across sectors and bank groups.
- To identify recurring seasonal patterns and any abrupt shifts or regime changes in lending behaviour.
- To compare growth, share, and volatility across bank groups; identify top performers and laggards.
- To evaluate rural–urban–metropolitan disparities in credit access and growth rates.

Data quality and reproducibility objectives

- To clean, normalize, and transform raw RBI tables into a tidy fact table with a full cleaning log and code/Power Query steps.

- To harmonize credit type labels, bank group names, and time formats so analyses are consistent and repeatable.

2. Source Description

The Dataset is maintained by ***India Data Portal (ISP)*** and is provided by **RBI(Primary source)** captures information on the credit extended by scheduled commercial banks in India to various population groups, bank groups and occupation groups. The Dataset covers various types of credit such as agricultural credits, personal loans, housing loans and loans to the industrial sectors etc.,

RBI releases information about the number of accounts, credit limit and amount outstanding categorised per population group, bank group and occupation under each district yearly.

2.1 Source Link

The Source link is

https://indiadataportal.com/p/reserve-bank-of-india/r/rbi-groupwise_credit_by_scbs-dt-yr-vvi

2.2 Time Line

The data set is created 2 years ago (2024) and last modified by 2025. The dataset contains credit limit data from 2019 to 2025. And the data is being updated yearly.

2.3 Domain

The domain of the data is Finance->Financial Inclusion.

3. Problem statement

- To map total credit by bank group across states, districts, and population segments (rural, urban, metropolitan) to identify geographic lending patterns.
- To measure year-wise changes in credit volumes to reveal long-term trends and recent shifts.
- To compare current credit levels with historical baselines to pinpoint regions and bank groups experiencing significant credit decline.
- To classify regions and bank-sector pairs into safe, moderate, critical, and over-exploited categories based on utilization and growth metrics.

- To decompose credit growth into volume and composition effects to understand whether growth is driven by sectoral shifts or overall expansion.
- To quantify sectoral composition (agriculture, industry, housing, personal) by region and bank group to detect structural changes in lending.
- To compute concentration and risk metrics (top-3 share, Herfindahl-Hirschman Index) to identify high-exposure bank groups and sectors.
- To produce an interactive dashboard with slicers and drilldowns for Year, Bank Group, Region, and Sector to support stakeholder exploration.

4. Attribute (Columns/Features) Details:

Attribute Name	Data Type	Description
Year	Number	Year of the credit
Region	Text	Region of the Bank
State	Text	Sate the bank located
District	Text	District of the band
Population Group	Text	Location type
Bank Group	Text	Type of bank
Occupation group	Text	Type of sector the loan was given
Occupation subgroup	Text	Subtype of the sector
Number of accounts	Number	No. of accounts lent money in particular Bank
Amount credit	Number	Amount of loan
Amount Outstanding	Number	Amount that to be paid by the sectors

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, scenarios and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing

6.1. Data Cleaning and Data Transformation

- The dataset has 10 Lakh plus records from the year 2013 to 2025, to ensure consistency, 2024- and 2025-year data only considered for analysis.
- The refined dataset consists of 348786 records.
- The Credit_ID Categorial column was added for readability.
- All categorial columns are cleaned and trimmed.
- All columns are assigned with appropriate datatypes.
- Validated for the null and blank values. Removed duplicates.
- Excel Functions used:
  - Clean, Trim and Proper.
- Calculated columns UtilizationRate is added for structured analysis.
  - ***UtilizationRate= OutStanding/CreditLimit***

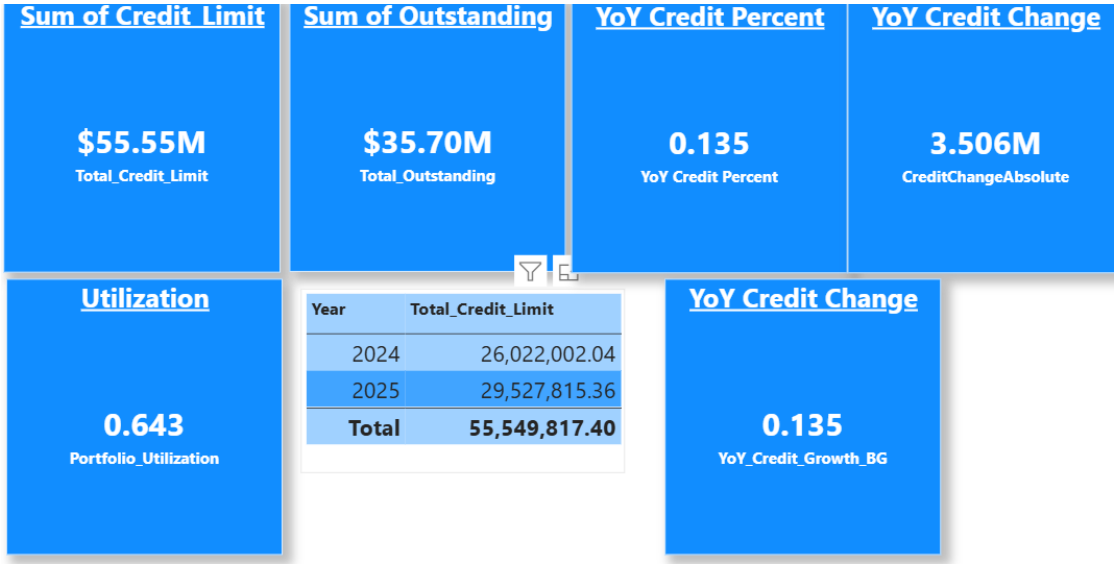
  - **Outstanding** → the current balance or exposure.
  - **CreditLimit** → the maximum sanctioned credit for that account.
  - **UtilizationRate** → shows how much of the available credit is being used.

7. Data Modelling and DAX

7.2. DAX

- Dax measures and calculated columns are created for the analysis as follows:
- ***Measures***

Total_Credit_Limit	Sum of lending amount over the years
Total_Outstanding	Sum of Outstanding over years
CreditChangeAbsolute	Difference of credit change over the years 2025 and 2024
CurrentCredit	Total of Credit of year 2025
PreviousCredit	Total of Credit of year 2024
Share Of Change (BankGroup or State)	**Share of Change** measure that shows how much each **BankGroup** or **State** contributes to the overall change in credit between two periods (e.g., YoY)
Top3_Share_Sector	Top 3 BankGroups by credit within sectors
HHI_Sector	purpose: concentration index (sum of squared shares) by sectors
YoY Credit Percent	Year-over-Year growth rate in credit, current year's credit exposure to the previous year's, expressed as a percentage change.
Utilization	Portfolio_Utilization = DIVIDE([Total_Outstanding], [Total_Credit_Limit], 0)



- ***Calculated columns***

  The following calculated columns are added.

  - YoY_Credit_BG

    This compares each BankGroup's credit exposure in the current year vs the previous year and expresses it as a percentage change.

```
YoY_Credit_Growth_BG =
VAR CurrYear = YEAR( MAX( Fact_Credit[Year] ) )
VAR PrevYear = CurrYear - 1

VAR CurrVal =
    CALCULATE(
        [Total_Credit_Limit],
        FILTER( ALL( Fact_Credit[Year] ), YEAR( Fact_Credit[Year] ) = CurrYear ),
        KEEPFILTERS( VALUES( Dim_Bank[Bank_Key] ) )
    )
VAR PrevVal =
    CALCULATE(
        [Total_Credit_Limit],
        FILTER( ALL( Fact_Credit[Year] ), YEAR( Fact_Credit[Year] ) = PrevYear ),
        KEEPFILTERS( VALUES( Dim_Bank[Bank_Key] ) )
    )

RETURN IF( ISBLANK( PrevVal ) || PrevVal = 0, BLANK(), DIVIDE( CurrVal - PrevVal, PrevVal, 0 ))
```

Bank_ID	BankGroup	PopulationType	Bank_Key	YoY_Credit_Growth_BG
B1	Rural	Public Sector Banks	Rural|Public	21.20%
B2	Rural	Private Sector Banks	Rural|Private Sector	8.17%
B3	Rural	Regional Rural Banks	Rural|Regional Rural	12.54%
B4	Rural	Small Finance Banks	Rural|Small Finance	32.07%
B5	Semi Urban	Public Sector Banks	Semi Urban|Public Sector	14.67%
B6	Semi Urban	Private Sector Banks	Semi Urban|Private Sector	13.71%
B7	Semi Urban	Foreign Banks	Semi Urban|Foreign	45.66%
B8	Semi Urban	Regional Rural Banks	Semi Urban|Regional Rural	9.72%
B9	Semi Urban	Small Finance Banks	Semi Urban|Small Finance	-4.42%
B10	Urban	Public Sector Banks	Urban|Public Sector	19.21%
B11	Urban	Private Sector Banks	Urban|Private Sector	13.93%
B12	Urban	Foreign Banks	Urban|Foreign	2.64%
B13	Urban	Regional Rural Banks	Urban|Regional Rural	13.90%
B14	Urban	Small Finance Banks	Urban|Small Finance	23.92%
B15	Metropolitan	Public Sector Banks	Metropolitan|Public Sector	13.31%
B16	Metropolitan	Private Sector Banks	Metropolitan|Private Sector	11.34%
B17	Metropolitan	Foreign Banks	Metropolitan|Foreign	8.14%
B18	Metropolitan	Regional Rural Banks	Metropolitan|Regional Rural	14.15%
B19	Metropolitan	Small Finance Banks	Metropolitan|Small Finance	14.30%
B20	Rural	Foreign Banks	Rural|Foreign	225.92%

  - ***High_UtilizationFlag***
  - ***HighUtilFlag = IF(Fact_Credit[Utilization Rate (Outstdg as % of Limit)] > 0.75, 1, 0)***

7.1. Data Modelling

- The original dataset was split into fact and dimension tables for structured analysis.
- Fact and dimension tables were created from the original dataset to support scenario modelling and descriptive analysis.
- Fact_credit contains following attributes

Credit_ID
Year
Reg_ID
Reg_Key
Sec_ID
Sec_Key
Bank_ID
Bank_Key
No. Of Accounts
Credit_Limit
UttilizationRate
ScenarioOutstanding
ScenarioUtilization
Outstanding

- From Fact_credit, Region, State and Districts and copied to separate sheet and normalized Region key ***Reg_Key*** is created by merging the Region,State and District content by delimiter.
    - ***TRIM([@Region]) & "\|" & TRIM([@State]) & "\|" & TRIM([@District])***
- ***Reg_ID*** is the text column, created with Flash fill of Excel by entering the first-row data as ***R1***.
- ***Fact_Credit*** Table as well as changed accordingly.
- The same ***Trim*** and ***Concatenation*** is used to create the ***Reg_Key*** column in the ***Fact_Credit*** Table.
- The ***Reg_ID*** column is created in ***Fact_Credit*** and ID is Pulled to the table by ***XLOOKUP.***
    - ***=XLOOKUP([@Reg_Key], Dim_Region[Reg_Key], Dim_Region[Reg_ID], "")***
- Now, the categorial columns, Region, State and Districts are deleted from the ***Fact_credit*** Table.
- In the same way, ***Dim_Bank*** and ***Dim_Sector*** tables are created and ID and Keys are mapped into the ***Fact_Credit***.
    - ***Dim_Region***

Reg_ID
Reg_Key
Region

State
District

- ***Dim_Sector***

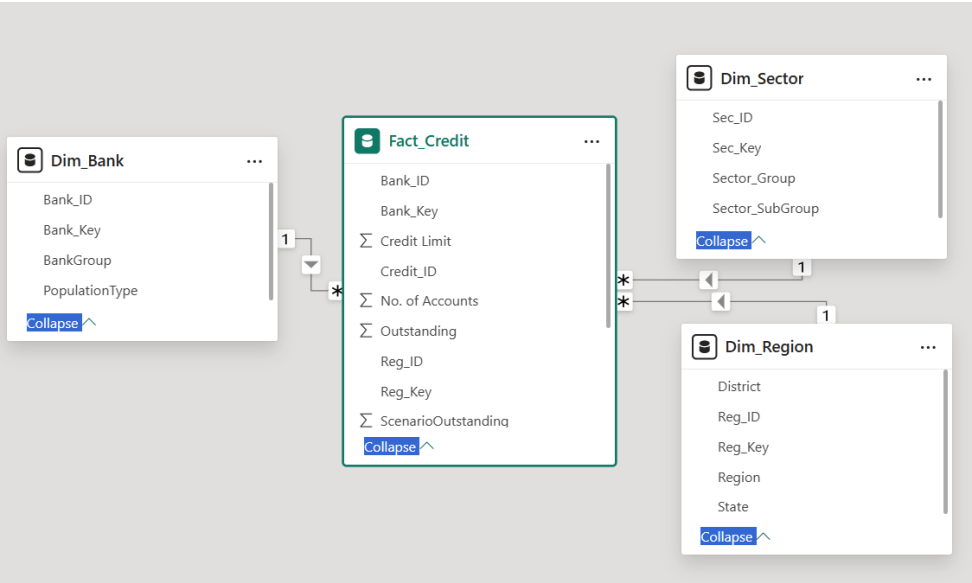
Sector_ID
Sec_Key
Sector
Sector_SubGrouop

- ***Dim_Bank***

Bank_ID
Bank_Key
Bank_group
Population_Type

Established relationships between the central **Fact_Credit** table and supporting dimension tables (**Dim_Bank**, **Dim_Sector**, **Dim_Region**).

- **Defined cardinality**:
  - One-to-many relationships from each dimension table to Fact_Credit.
  - Each Bank, Sector, and Region can have multiple credit records in Fact_Credit.
- **Created lookup tables** where necessary:
  - **Dim_Bank** provides descriptive attributes such as BankGroup and PopulationType.
  - **Dim_Sector** organizes Sector_Group and Sector_SubGroup classifications.
  - **Dim_Region** supplies geographic context (District, Region, State).

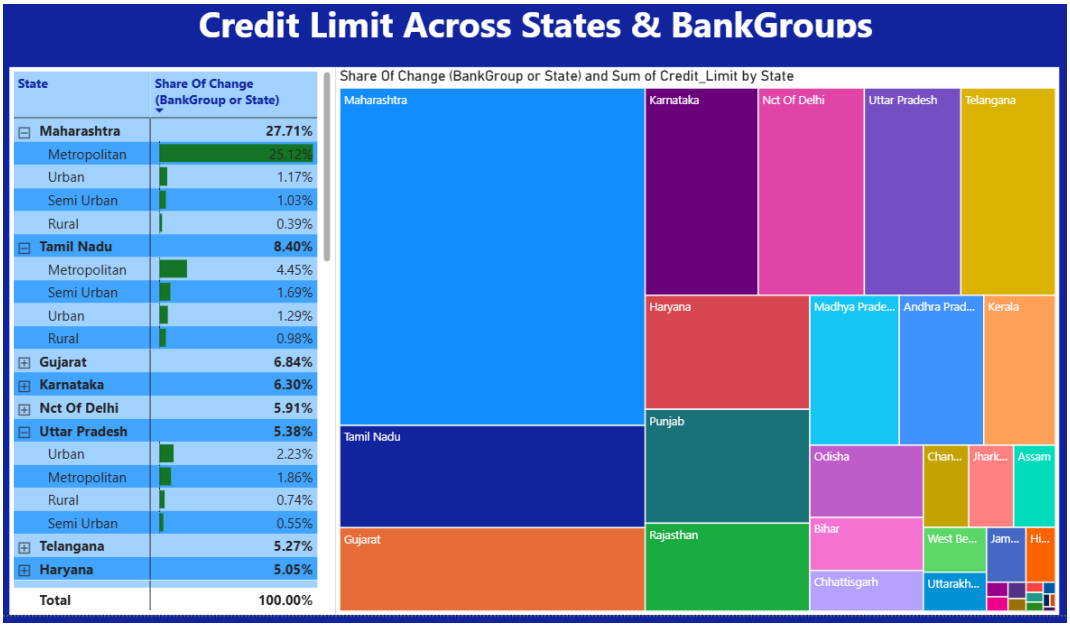


- **Keys and foreign keys**:
  - Fact_Credit contains Bank_ID, Sec_ID, and Reg_ID as foreign keys.
  - Dimension tables hold corresponding primary keys (Bank_Key, Sec_Key, Reg_Key) for referential integrity.
- **Schema design**: Implemented a **star schema** with Fact_Credit at the center and dimensions surrounding it, enabling efficient slicing, aggregation, and scenario analysis.
- **Purpose**: Supports descriptive statistics, scenario modeling, and geographic/institutional breakdowns in PivotTables and Power BI dashboards.

8. Analysis and Visualizations

8.1. Credit Across States and Bank Groups

- This visual is showing the credit exposure of bank groups such as Metropolitan, Rural, Urban and Semi-Urban throughout the states of India.



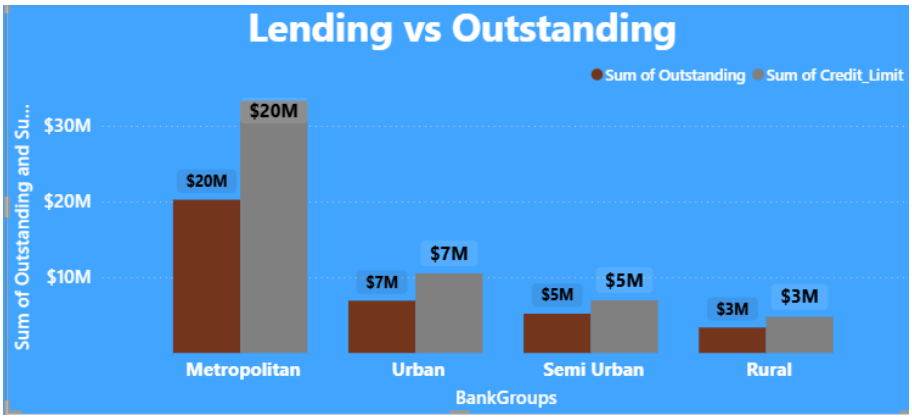
State	Share Of Change (BankGroup or State)
Maharashtra	27.71%
Metropolitan	25.12%
Urban	1.17%
Semi Urban	1.03%
Rural	0.39%
Tamil Nadu	8.40%
Metropolitan	4.45%
Semi Urban	1.69%
Urban	1.29%
Rural	0.98%
Gujarat	6.84%
Karnataka	6.30%
Nct Of Delhi	5.91%
Uttar Pradesh	5.38%
Urban	2.23%
Metropolitan	1.86%
Rural	0.74%
Semi Urban	0.55%
Telangana	5.27%
Haryana	5.05%
Total	100.00%

Insights

- **Maharashtra (27.7%)**, **Tamil Nadu (10.3%)**, and **Gujarat (7.0%)** together account for roughly **45%** of the portfolio's recent share-of-change.
- Metro and urban centres are the primary drivers of that growth, with metropolitan contributions notably large in Maharashtra, Tamil Nadu, and Uttar Pradesh.
- The portfolio's recent growth is **highly concentrated in a few states and metropolitan centers**; immediate priorities are to quantify counterparty concentration within those states, stress-test top contributors, and pursue targeted geographic diversification to reduce systemic vulnerability.

8.2. Lending versus Outstanding over BankGroups

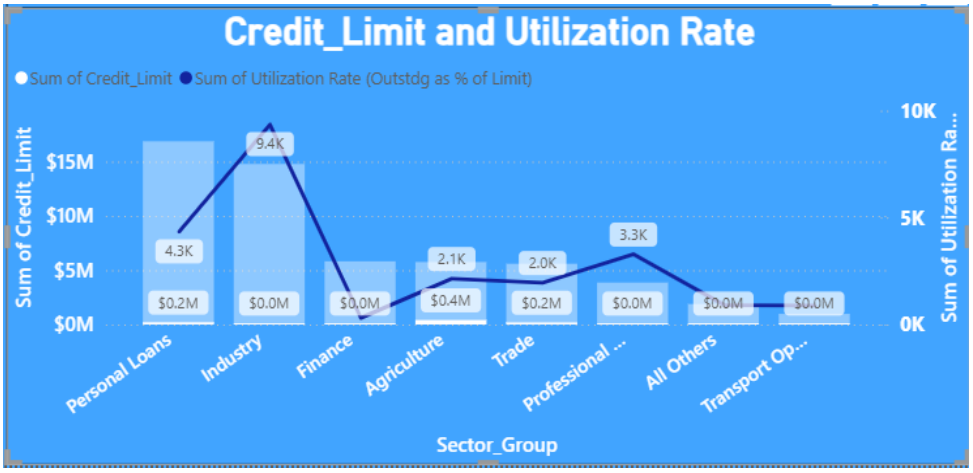
- The chart shows **identical values for Sum of Outstanding and Sum of Credit Limit** across Metropolitan, Urban, Semi Urban, and Rural — Metropolitan dominates (~20M), then Urban (~7M), Semi Urban (~5M), Rural (~3M). That pattern implies either **very high utilization (≈100%)**



Insights:

- **High rollover risk** — renewals or limit increases are required to maintain lending; any refusal to renew will immediately reduce active exposure.
- **Concentration**: Metropolitan accounts for the majority of portfolio exposure.
- **Metropolitan concentration** — Metropolitan holds the largest exposure (~20M), roughly **50%+** of total portfolio shown, making the portfolio highly sensitive to metro-level shocks.
- **Uniformity across segments** — Similar utilization across Urban, Semi Urban, and Rural suggests systemic behavior (policy or market) rather than isolated pockets of drawdown.

8.3. Credit Exposure and Utilization over Sectors



Insights:

- **Industry** is the dominant exposure with the largest credit limit and the highest utilization signal;
- **Personal Loans** and **Agriculture** show meaningful exposure but much lower utilization in absolute terms.

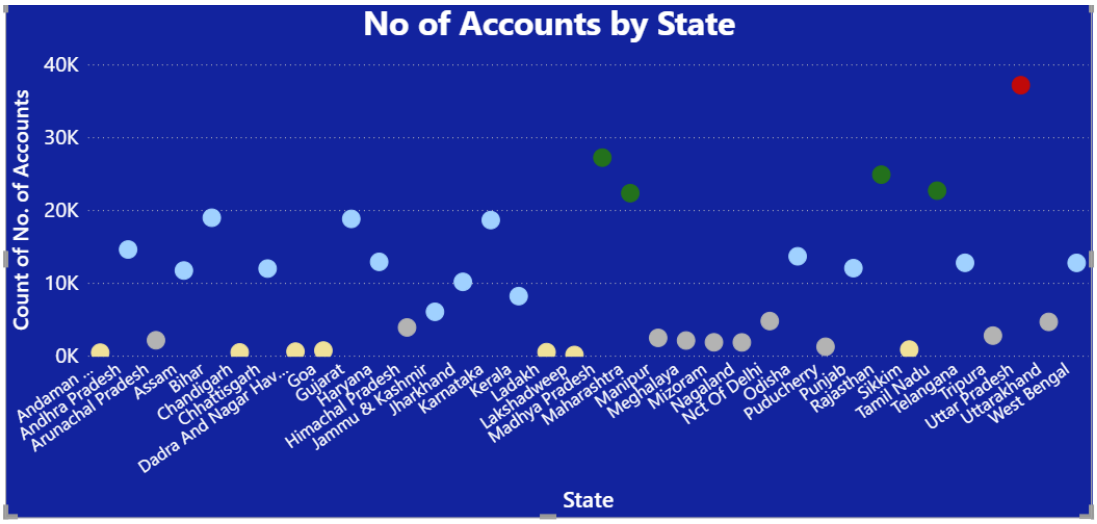
- Concentration risk: ***Industry and Personal Loans*** together represent the largest share of sanctioned limits; shocks in these sectors will move portfolio metrics materially.
- Hotspots with limited scale: ***Professional Services and Transport*** show higher utilization rates despite smaller absolute exposure; these sectors can be early indicators of stress that may escalate if left unchecked.

8.4. Matrix for YoY Credit Percentage vs Regions/BankGroups

Region	YoY Direction	YoY Credit Percent
Western Region	**Increase**	**0.14**
Urban	Increase	0.14
Semi Urban	Increase	0.07
Rural	Increase	0.13
Metropolitan	Increase	0.15
Southern Region	**Increase**	**0.13**
Urban	Increase	0.14
Semi Urban	Increase	0.15
Rural	Increase	0.14
Metropolitan	Increase	0.11
Northern Region	**Increase**	**0.14**
Urban	Increase	0.17
Semi Urban	Increase	0.17
Rural	Increase	0.28
Metropolitan	Increase	0.10
North Eastern Region	**Increase**	**0.16**
Urban	Increase	0.19
Semi Urban	Increase	0.12
Rural	Increase	0.14
Metropolitan	No prior year	
Eastern Region	**Increase**	**0.10**
Urban	Increase	0.19
Semi Urban	Increase	0.17
Rural	Increase	0.11
Metropolitan	Increase	0.00
Central Region	**Increase**	**0.15**
Urban	Increase	0.19
Total	**Increase**	**0.13**

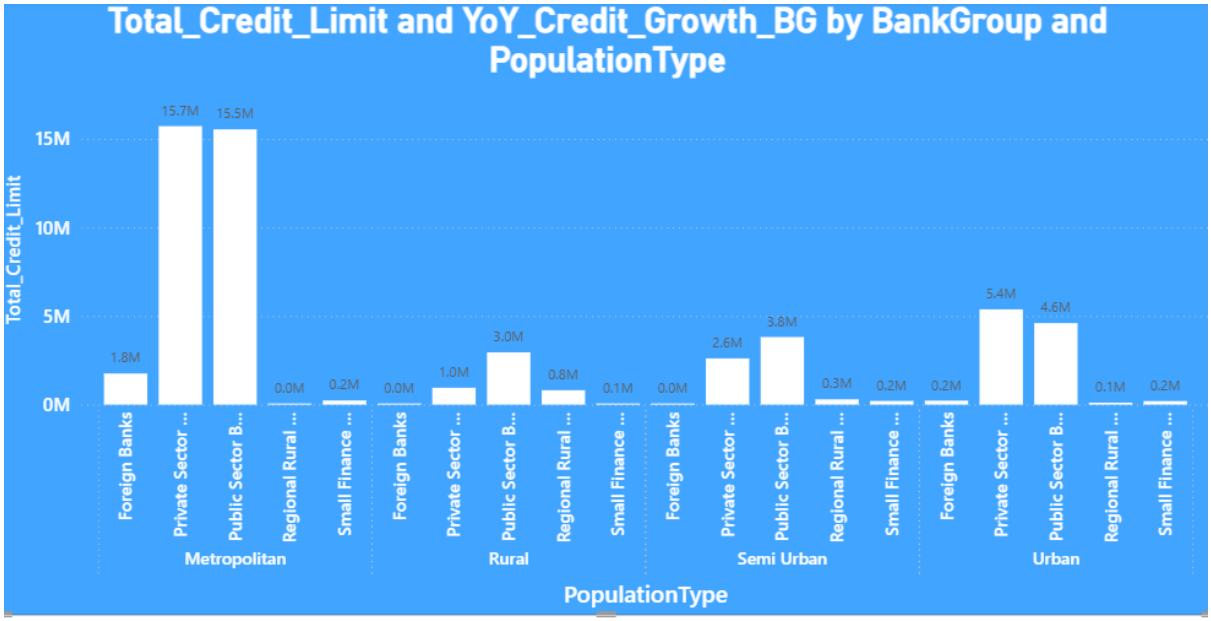
- The visual shows broad YoY credit growth across all regions, with an overall YoY credit increase of 13%.
- Growth is fairly uniform across metropolitan, urban, semi-urban and rural subregions, but clear hotspots and data gaps stand out most notably very strong rural growth in the Northern region and a missing/zero metropolitan value in the Eastern region.

8.5. State-Level Account Concentration and Distribution



- **Top states dominate** — a small number of states hold the largest share of accounts (one state near 40K), creating geographic concentration in customer base.
- **Wide dispersion** — many states have low account counts, indicating uneven market penetration and potential untapped opportunity.
- **Clustered mid-tier group** — several states cluster around ~20K accounts, forming a middle band that drives steady volume but not dominance.
- **Long tail of low-count states** — numerous states with very low counts increase operational overhead per account and suggest inefficient coverage.
- **Outlier risk** — the single high-count outlier increases dependency on that state for growth and collections performance.

8.6. YoY Credit Growth

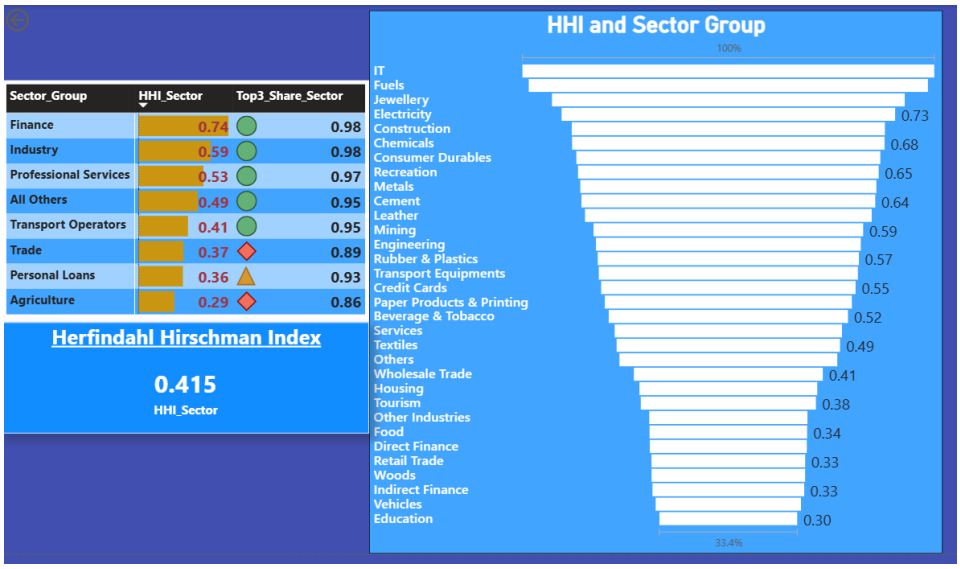


- The chart shows Private and Public Sector Banks dominate metropolitan and urban credit, while Public Sector Banks and Regional Rural Banks lead in rural areas. Metropolitan exposure is the largest concentration overall, creating

geographic and counterparty concentration that requires targeted monitoring and stress testing.

- ***Insights:***
- **Metropolitan**: **Private Sector Banks ≈ 15.7M**, **Public Sector Banks ≈ 15.5M**. These two bank groups together hold the vast majority of metro credit.
- **Urban**: **Foreign Banks ≈ 5.4M**, **Private Sector Banks ≈ 4.6M** — foreign banks have a surprisingly large urban footprint.
- **Semi Urban**: **Foreign Banks ≈ 2.6M**, **Private Sector Banks ≈ 3.8M.** semi-urban lending is split between foreign and private banks.
- **Rural**: **Public Sector Banks ≈ 3.0M**, **Regional Rural Banks ≈ 0.8M**. Rural credit is concentrated in public and regional rural banks.
- **Small Finance Banks** show minimal aggregate limits across all population types (≤0.2M), indicating limited scale in this dataset.
- **Diversify origination** into semi-urban and rural segments where credit per bank is lower, using digital channels or partnerships to scale cost-effectively.

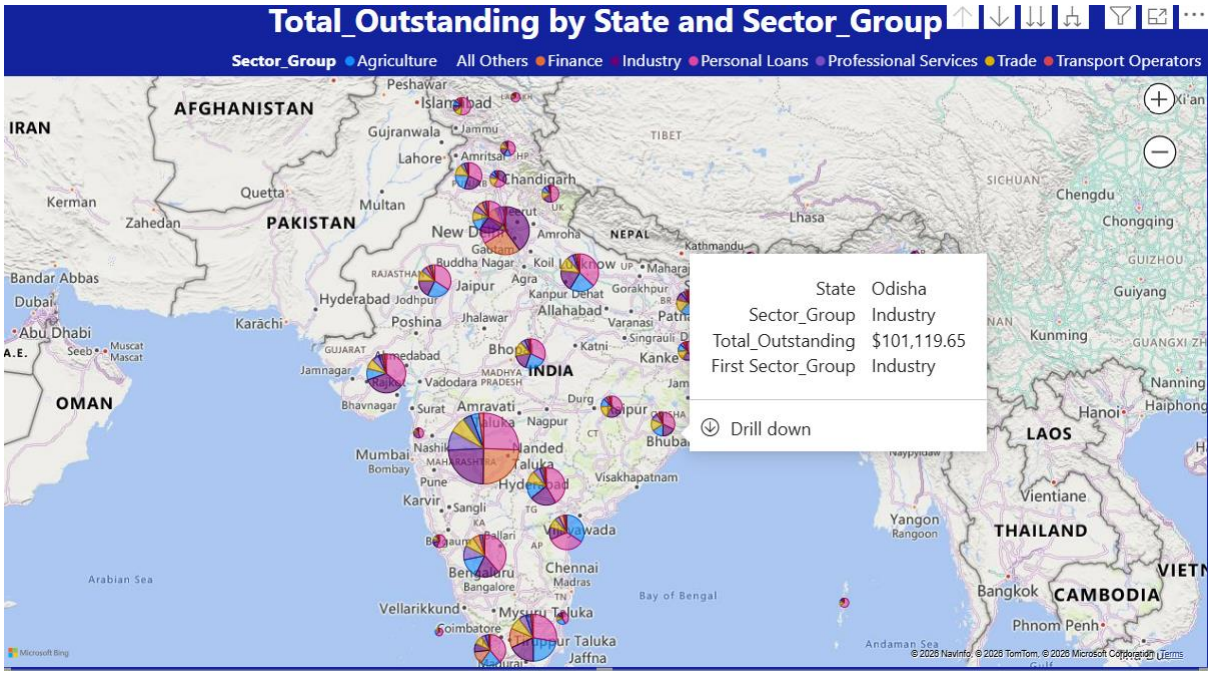
8.6. HHI (Herfindahl–Hirschman Index)



Sector_Group	HHI_Sector	Top3_Share_Sector
Finance	0.74	0.98
Industry	0.59	0.98
Professional Services	0.53	0.97
All Others	0.49	0.95
Transport Operators	0.41	0.95
Trade	0.37	0.89
Personal Loans	0.36	0.93
Agriculture	0.29	0.86

- The **Herfindahl–Hirschman Index (HHI)** is a standard measure of market concentration used in economics and risk analysis.
- The sector landscape is **highly concentrated**. A few sectors (notably **IT, Fuels, Jewellery, Electricity, Construction, Chemicals, Industry, Finance**) show very high HHI and **Top-3 shares near or above 90%**, while sectors like **Agriculture, Trade, Personal Loans** are comparatively more distributed. The portfolio-level HHI (~**0.42**) signals **moderate to high concentration**.
- This shows **sector-level concentration** using HHI and the **Top-3 share**. **Finance, Industry, and Professional Services** are the most concentrated sectors, while **Agriculture, Trade, and Personal Loans** are relatively more distributed.

8.7. Outstanding by Sectors



- ***Insights***
- The **outstanding portfolio is sector-skewed**: a handful of sectors (notably **Industry, Finance, Personal Loans, and Agriculture**) drive most of the outstanding balances across states.
- Geographic distribution is uneven — a few states show large, multi-sector pies while many states have small, narrow exposures — creating both **concentration risk** and clear opportunities for targeted growth or remediation.
- **Metro hubs concentrate multi-sector exposure** — states with big urban centers show large, diversified pies but are still dominated by Industry/Finance.
- **Rural and semi-urban pies are sector-specific** — Agriculture and Regional Trade dominate in some states, making those states vulnerable to local shocks (monsoon, commodity prices).
- **Single-sector dominance in some states** suggests **single-point failure risk** if a major borrower or industry cluster weakens.
- **Long tail of small states**: many states have low outstanding; these are opportunities for low-cost digital origination or partnerships to diversify.

9. Insights & Conclusions

- Since the dataset is huge, Statistical Analysis is done by pivot table.

Row Labels	Sum of Amount Out	Average of Amount Ou	Min of Amount (	Max of Amount Ou	StdDev of Amount Outstanding
Central Region	3613882.53	44.6130798	0	26916.2055	355.4473098
Eastern Region	2717878.511	47.83565677	0	36438.4612	428.4203873
North Eastern Region	435791.6878	17.53406646	0	4983.1911	114.4211368
Northern Region	7416875.951	113.2900952	0	234201.4501	1775.338444
Southern Region	10799468.62	138.0759022	0	128261.1973	1461.274873
Western Region	10716662.45	252.590625	0	576157.3039	4837.882433
Grand Total	**35700559.74**	**102.356924**	**0**	**576157.3039**	**1995.601825**

9.1. Central Tendency

- **Grand Total Average Outstanding per account**: ~102.36
- **Highest Average**: Western Region (~252.59) → indicates larger account sizes.
- **Lowest Average**: North Eastern Region (~17.53) → smaller account sizes.

  Western Region has accounts that are, on average, ~15× larger than North Eastern.

9.2. Spread

StdDev (Volatility):

  - Western Region: ~4,837 → very high variability in exposures.
  - Northern Region: ~1,775 → also highly dispersed.
  - North Eastern Region: ~114 → relatively stable, low variability.

- **Range (Max – Min)**:
- Western Region: 576,157 – 0 → extremely wide spread.
- Northern Region: 234,201 – 0 → concentrated large exposures.
- Other regions show smaller ranges.
  - Western and Northern regions carry **concentration risk** (few very large accounts drive totals).

9.3. Distribution (Geographic Lending Patterns)

- **Total Exposure (Sum Outstanding)**:
  - Southern Region: ~10,799,469 (largest share).
  - Western Region: ~10,716,662 (almost equal to Southern).
  - Northern Region: ~7,416,876.
  - Central + Eastern combined: ~6,331,761.
  - North Eastern: ~435,792 (smallest share).
  - **% of Grand Total (35,700,560)**:
  - Lending is **heavily concentrated in Southern + Western (≈60%)**, while North Eastern is negligible.

9.3. Key Insights

  - **Top Performers (High exposure, moderate volatility)**: Southern Region → large exposure, but StdDev lower than Western.

- **Risky Growth (High exposure, high volatility)**: Western Region → largest accounts, widest spread.
- **Stable but Stagnant**: North Eastern → small exposure, low volatility.
- **Concentration Risk**: Northern Region → very high max exposure (234,201), suggesting dependency on a few large borrowers.

9.4. Analysis By Pivot Tables

9.4.1 CreditLimit_By_Bank_And_Sector

Sum of Credit Limit (in Rs Crores) Row Labels	Column Labels Agriculture	All Others	Finance	Industry	Personal Loans	Professional S	Trade	Transport Operat	Grand Total
2024									
Metropolitan|Foreign	2,473.69	26,317.35	138,634.73	354,473.88	84,502.98	135,595.73	91,335.57	10,961.65	844,295.60
Metropolitan|Private Sector	135,677.92	349,370.26	645,348.18	2,023,187.32	2,610,052.37	790,044.59	748,545.02	135,932.01	7,438,157.67
Metropolitan|Public Sector	196,668.07	208,479.69	1,595,284.42	2,820,470.98	1,353,967.21	382,088.69	607,840.55	119,361.55	7,284,161.16
Metropolitan|Regional Rural	3,098.94	839.77	348.79	627.25	7,504.36	588.72	1,466.71	114.62	14,589.16
Metropolitan|Small Finance	13,024.51	3,975.94	10,423.07	11,693.52	27,206.21	14,488.36	23,395.13	5,890.62	110,097.36
Rural|Foreign	0.50	118.95	12.10	113.22	51.20	0.64	65.63		362.24
Rural|Private Sector	167,002.26	11,223.84	3,883.93	74,538.90	122,957.81	20,329.64	44,248.79	13,061.24	457,246.40
Rural|Public	586,241.44	5,673.70	26,499.73	53,353.43	385,824.70	26,945.75	243,792.31	6,436.24	1,334,767.30
Rural|Regional Rural	265,529.33	22,590.07	20,556.64	7,506.71	40,195.56	6,030.90	19,428.03	1,052.94	382,890.19
Rural|Small Finance	15,011.30	1,888.61	53.97	1,822.59	2,267.89	780.18	3,141.72	464.81	25,431.06
Semi Urban|Foreign	3.73	1,419.48	75.03	5,630.74	5,632.23	270.09	1,113.56	28.16	14,173.02
Semi Urban|Private Sector	239,807.00	58,201.98	10,256.11	223,986.27	456,058.24	60,825.96	129,935.05	42,084.41	1,221,155.01
Semi Urban|Public Sector	558,746.87	9,136.27	20,188.31	202,823.81	784,252.75	52,770.53	138,959.81	11,072.59	1,777,950.94
Semi Urban|Regional Rural	79,577.15	8,070.58	7,587.65	4,263.11	31,438.34	2,753.45	8,566.90	648.95	142,906.13
Semi Urban|Small Finance	46,345.64	4,163.77	159.12	8,331.93	11,434.29	12,467.04	18,380.99	5,907.22	107,190.01
Urban|Foreign	142.61	3,432.98	12,187.72	64,245.60	8,335.36	10,351.07	9,578.43	3,362.23	111,636.01
Urban|Private Sector	156,748.79	117,644.00	161,895.12	606,048.51	965,981.40	166,204.01	256,638.62	83,607.37	2,514,767.82
Urban|Public Sector	196,378.81	32,817.62	96,459.52	468,662.90	911,761.37	123,449.50	250,125.23	19,411.16	2,099,066.11
Urban|Regional Rural	13,354.69	5,453.85	2,243.92	2,457.15	21,412.76	1,902.52	4,164.93	273.28	51,263.10
Urban|Small Finance	26,205.13	4,098.22	1,783.51	7,437.10	16,593.60	7,541.06	19,960.40	6,276.73	89,895.76
2025									
Metropolitan|Foreign	78,235.86	14,832.52	141,282.29	350,871.90	97,995.88	127,543.05	90,852.67	11,380.30	912,994.48
Metropolitan|Private Sector	157,041.70	433,371.85	592,309.23	2,342,160.63	2,939,313.53	825,625.91	840,390.93	151,213.33	8,281,427.11
Metropolitan|Public Sector	168,949.39	214,472.44	1,872,271.87	3,148,464.46	1,634,807.62	445,955.17	649,801.77	119,163.31	8,253,886.03
Metropolitan|Regional Rural	2,919.86	577.27	341.03	1,214.20	9,242.49	532.31	1,686.11	140.35	16,653.62
Metropolitan|Small Finance	11,662.67	7,218.54	8,064.71	11,458.81	29,357.82	25,297.95	25,512.60	7,273.46	125,846.57

- **Key Insights**
- **Overall Growth:**
  - Total Credit Limit increased from ~₹47.4 lakh crores in 2024 to ~₹55.5 lakh crores in 2025 (≈17% growth).
- **Sectoral Leaders:**
  - **Industry** and **Personal Loans** dominate, together accounting for ~57% of total credit limits in 2025.
  - **Finance** and **Trade** are mid-tier contributors, while **Agriculture** remains relatively small.
- **Institutional Patterns:**
  - **Public Sector Banks** hold the largest share across regions, especially in Rural and Semi-Urban areas.
  - **Private Sector Banks** show strong growth in Metropolitan and Urban regions, surpassing Public Sector in some categories.
  - **Small Finance Banks** remain low players but show steady expansion.
- **Regional Distribution:**

- **Metropolitan regions** account for the majority of credit limits, with Private and Public Sector banks leading.
- **Urban regions** show strong growth, especially in Private Sector lending.
- **Rural regions** are dominated by Public Sector banks, reflecting government-driven credit penetration.
- **Semi-Urban regions** contribute significantly, with Public Sector banks maintaining leadership.

- **Foreign Banks:**
- Their share is small but concentrated in Metropolitan and Urban regions, mainly in Industry

9.4.2. Exposure By Region and Bank

- **Overall Scale:**
  - Total Outstanding = ₹35.7 lakh crores.
  - Metropolitan areas dominate (₹20.2 lakh crores, ~57% of total).
- **Regional Leaders:**
  - **Western Region** is the largest contributor (₹10.6 lakh crores), driven mainly by Maharashtra (~₹8.5 lakh crores).
  - **Southern Region** follows (~₹11.7 lakh crores), with Tamil Nadu (~₹3.3 lakh crores) and Karnataka (~₹2.6 lakh crores) as major hubs.
  - **Northern Region** is significant (~₹7.2 lakh crores), heavily concentrated in Delhi (~₹3.5 lakh crores).
  - **Eastern and North Eastern Regions** are smaller contributors (~₹2.7 lakh crores combined).

Sum of Amount Outstanding (in Rs Crores)	Column Labels				
Row Labels	Metropolitan	Rural	Semi Urban	Urban	Grand Total
Central Region					
Chhattisgarh	185,691.20	42,850.92	70,446.01	117,682.06	416,670.20
Madhya Pradesh	481,880.78	131,719.55	230,407.08	195,339.70	1,039,347.11
Uttar Pradesh	646,364.14	385,940.38	285,982.45	637,796.23	1,956,083.20
Uttarakhand		37,911.82	42,291.69	121,578.51	201,782.02
Eastern Region					
Andaman & Nicobar Islands		1,684.38	1,398.02	5,533.08	8,615.48
Bihar	147,589.28	143,698.62	142,850.75	147,089.38	581,228.02
Jharkhand	99,767.12	53,483.79	55,308.19	73,473.72	282,032.82
Odisha		122,551.78	131,164.17	310,812.35	564,528.31
Sikkim		4,220.07	1,365.27	8,134.01	13,719.36
West Bengal	678,571.21	194,481.39	126,976.10	267,725.83	1,267,754.52
North Eastern Region					
Arunachal Pradesh		5,883.23	13,841.61		19,724.84
Assam		68,094.20	82,760.06	137,262.46	288,116.72
Manipur		8,046.95	5,409.09	12,086.01	25,542.06
Meghalaya		10,369.64	5,588.69	13,730.25	29,688.58
Mizoram		2,962.35	4,385.66	9,736.84	17,084.85
Nagaland		4,089.24	9,083.67	5,918.86	19,091.76
Tripura		13,063.05	9,096.62	14,383.21	36,542.88
Northern Region					
Chandigarh		836.77	1,807.54	178,875.61	181,519.92
Haryana	105,210.90	100,074.81	166,815.07	887,490.24	1,259,591.03
Himachal Pradesh		68,779.91	38,734.42	11,489.44	119,003.77
Jammu & Kashmir	29,574.71	80,070.07	67,699.46	43,565.51	220,909.75
Ladakh		3,462.14	4,879.52		8,341.66
Nct Of Delhi	3,395,019.59	11,469.31	21,367.75	98,822.07	3,526,678.72

- **State-Level Concentration:**
  - **Maharashtra, Tamil Nadu, Karnataka, Delhi, and Uttar Pradesh** together account for the majority of national outstanding.
  - Kerala shows a unique pattern: very high Semi-Urban and Rural exposure (~₹1.2 lakh crores total).
- **Urban vs Rural Split:**
  - Metropolitan + Urban together = ~77% of total outstanding.
  - Rural + Semi-Urban = ~23%, highlighting concentration in large cities.
- **Risk Concentration:**
- Heavy reliance on a few states (Maharashtra, Tamil Nadu, Karnataka, Delhi, UP).
- Smaller states/UTs (e.g., Sikkim, Lakshadweep, Goa) contribute marginally.
- Summary: **Outstanding credit is highly concentrated in metropolitan hubs and a handful of large states, with Maharashtra and Tamil Nadu leading. Rural and semi-urban exposure is meaningful in Kerala and parts of Eastern India, but overall, the portfolio is city-centric.**

9.4.2. Analysis by Scenario Manager

Scenario Analysis Insights

- Base Scenario (GrowthFactor = 1.10, UtilizationAdj = 0.00):
  - Assumes steady growth with no change in utilization.

- ➢ Metropolitan and Urban segments remain the largest contributors, reflecting strong credit demand in city hubs.
- ➢ Rural and Semi-Urban segments show moderate exposure, but utilization is already high, signaling saturation risk.

- Optimistic Scenario (GrowthFactor = 1.00, UtilizationAdj = –0.02):
  - ➢ Growth is flat, but utilization decreases slightly.
  - ➢ This scenario suggests improved efficiency: borrowers are using less of their sanctioned limits, reducing risk.
  - ➢ Metropolitan and Urban segments benefit most, as lower utilization balances their high exposure.
  - ➢ Rural and Semi-Urban segments still show very high utilization, indicating limited room for expansion.
- Stress Scenario (GrowthFactor = 0.90, UtilizationAdj = +0.05):
- Growth slows, while utilization rises.
- Rural and Semi-Urban segments become particularly vulnerable, as their utilization rates spike further.
- Metropolitan exposure remains large, but stress conditions amplify volatility.
- Summary
- Metropolitan & Urban → dominate exposure; optimistic scenario reduces risk slightly, but stress conditions could magnify volatility.
- Rural & Semi-Urban → already high utilization; stress scenario pushes them into riskier territory, while optimistic scenario offers little relief.
- Portfolio Sensitivity → utilization is the critical lever: small adjustments (+/–2–5%) significantly change risk outlook.
- Strategic Implication → diversify exposure away from high-utilization rural/semi-urban accounts, and monitor large metropolitan borrowers under stress conditions.
- **Scenario Sheet:**

Optimistic

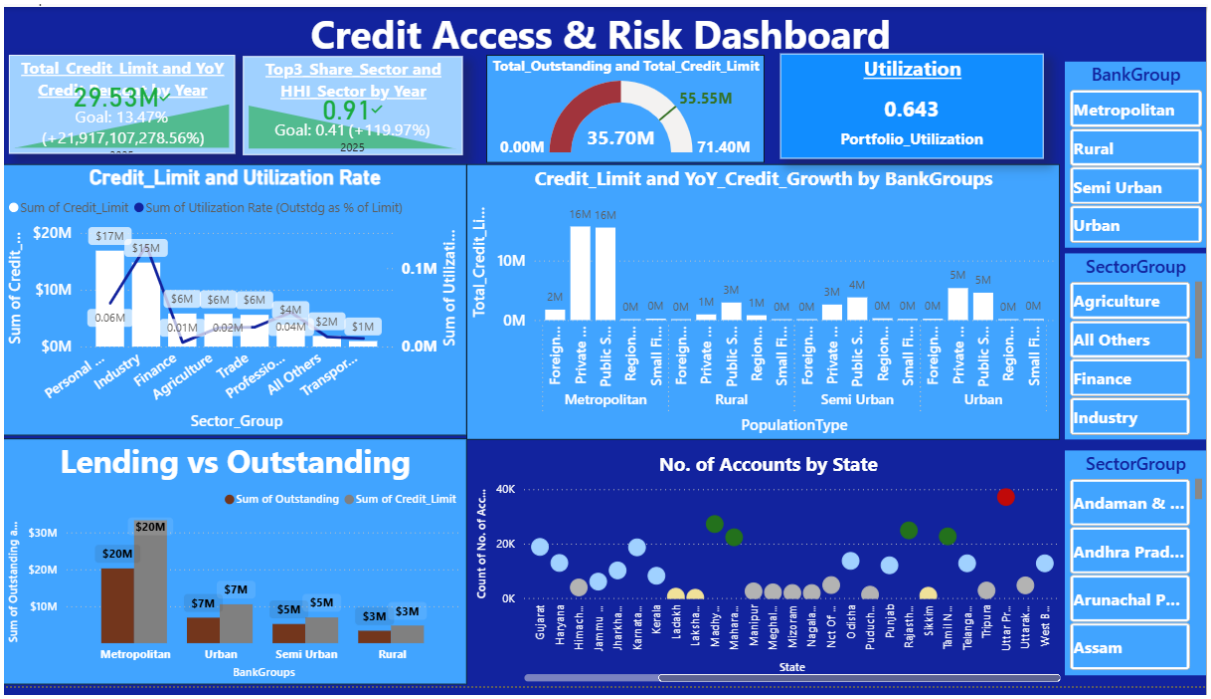
B	C	D	E	F	G
	Scenario Lists				
Scenarios	Optimistic				
			Scenarios	GrowthFactor	UtilizationRate
			Base	1.10	0.00
			Optimistic	1.00	-0.02
			Stress	0.90	0.05

Row Labels	Sum of Avg_credit_Limit_per_AC	Sum of Avg_Outstanding_Per_AC	Sum of Utilization Rate (Outstdg as % of Limit)
Metropolitan	36923.9692	23530.73148	11214.3542
Rural	23296.84659	12160.73295	86037.46984
Semi Urban	35325.29849	21247.28435	98528.98635
Urban	47570.43397	28979.00428	56974.31468
Grand Total	**143116.5483**	**85917.75305**	**252755.1251**

- **Calculated column in Fact_credit**

ScenarioUtilization	ScenarioOutstanding
-0.02	0.7952
-0.02	0.02
-0.02	1.0523
-0.02	0.0144
-0.02	0.015
-0.02	0.0087
-0.02	0.1212
-0.02	0.1012
-0.02	0.3458
-0.02	0.0125
-0.02	1.9085
-0.02	0.0029
-0.02	0.1006
-0.02	4.0145
-0.02	0.0177
-0.02	0.5135
-0.02	0.008
-0.02	5.0596
-0.02	0.9607
-0.02	0.3019
-0.02	0.0123
-0.02	0.2275
-0.02	0.7265
-0.02	0.1154
-0.02	0.1409
-0.02	0.0141
-0.02	0.1424
-0.02	0.0093
-0.02	0.9605
-0.02	0.0131
-0.02	0.1311

9.5. Overall Summary



The dashboard shows a **portfolio with strong growth but elevated concentration and utilization risk**. Total sanctioned capacity has expanded materially, utilization is high, and a few sectors, states, and BankGroups dominate

exposure — so the immediate priority is to validate drivers, quantify concentration, and run targeted stress tests.

- **Portfolio Utilization: ~0.64** (64%) significant drawdown of sanctioned lines.
- **Portfolio HHI: ~0.42** moderate-to-high concentration at sector level.
- **Top-3 Share Sector:** the three players ***Industry, Personal Loans and Finance*** dominate many sectors.
- **Segment concentration***: **Metropolitan and Industry/Finance** sectors* are the largest buckets; ***Personal Loans and Agriculture*** are material but smaller.

9.6. Diagnosis Summary · Operational View · Forecast

- Rapid sanctioned growth with **high utilization (~64%)** and **elevated concentration (HHI ≈ 0.42)**; outstanding is heavily drawn and metro/Industry/Finance are the primary hotspots.
- That makes the portfolio fragile—there's little spare capacity and too much reliance on a few players.
- If unchanged, expect **rising delinquencies and higher volatility within 3–6 months**; a 20–30% non-renewal or shock to top contributors would produce material portfolio stress.

10. Conclusion

The portfolio is **large, growing, and heavily concentrated in metropolitan hubs and a few states**, with **Industry and Personal Loans as the main sectors**. While growth is strong, the **volatility and concentration risk in Western/Northern regions** and the **urban bias of lending** highlight the need for diversification into rural/semi-urban markets and broader sectoral balance.